

Endocrinology

Outpatient Type 2 Diabetes Mellitus

SCREENING ([Diab Care 2026:49:S35](#))

ADA recs: Begin ≥ 35 years, repeat q3y if normal. Repeat q1y if pre-DM/wt gain. Higher risk = 1) BMI ≥ 25 (≥ 23 in Asian-Americans) + RF (1st degree relative with T2DM, nonwhite, history of CVD, HTN, HDL < 35 , triglycerides > 250 , PCOS, sedentary lifestyle, acanthosis nigricans, MASLD) or 2) History of gestational DM (screen q1-3y) or pancreatitis or 3) HIV prior to starting/switching ART. ([ADA 2026](#)).

PRE-DIABETES ([Diab Care 2026:49:S35](#))

- **Diagnosis:** A1c 5.7-6.4%; fasting plasma glucose (FPG) 100-125; or 75g OGTT w/ 2h glucose 140-199
- **Monitoring:** A1c at least q1y; if A1c 6-6.4%, screen q6mo; at initiation of second-gen antipsychotics & repeat 12-16 weeks after; prior to starting HIV ART, switching ART, & 3-6 months after.
- **Treatment:** **lifestyle** Δ most effective; **metformin** also effective, esp. if BMI ≥ 35 , age < 60 , or GDM hx ([Cochrane Rev 2019](#))

DIABETES ([Diab Care 2025:48:S27](#))

- **Diagnosis:** A1c $\geq 6.5\%$; FPG ≥ 126 ; 75g OGTT with 2h glucose ≥ 200 ; or random BG ≥ 200 & symptoms. Unless diagnosis is made by symptoms & random glucose > 200 , **confirm with repeat or additional test at the same time or promptly after.**
 - For **T1DM**, check TSH at dx and q1y. Screen for thyroid dz and celiac at dx, and again if signs/sx develop. ([Diab Care 2026:49:S63](#))
 - Use plasma blood glucose criteria and not A1C if high RBC turnover: sickle cell disease, 2nd/3rd trimester of pregnancy, $\downarrow$ G6PD, HD, HIV, recent blood loss/transfusion, EPO tx.
 - A1c less reliable post-partum, with certain HIV drugs, thalassemia and sickle cell (high cell turnover), and anemia ([UTD list](#))
- Tx: goal A1c $< 7\%$; can liberalize to $< 8-8.5\%$ if life expectancy ≤ 10 years or high risk for hypoglycemia (e.g., ESRD)

Healthcare Maintenance for Diabetic Patients (Diab Care 2026:49:S61-89)	
Every visit	• Review blood sugar log: goal AM FPG 80-130, postprandial (1-2h) < 180; eval for hypoglycemia frequency, causes, and timing • Review medication regimen and medication-taking behavior. Screen for tobacco, EtOH, and substance use. Counsel. • Blood pressure: goal $< 130/80$. ACEi/ARB first line. Can target $< 120/80$ in pts w/ multiple CV RFs (if it can be safely attained). • Weight, BMI: weight center referral if BMI ≥ 40 or ≥ 35 with poor control; nutrition referral for all (JAMA 2023) • Foot exam (inspect skin, joints, pulses, sensation) esp. if known neuropathy or PVD; ABIs/vascular referral if PVD • Skin exam (acanthosis nigricans, insulin injection/insertion sites, lipodystrophy)
Q3-6mo	• A1c q6mo if controlled and stable regimen; q3mo if $\uparrow$ target, recent tx change, hypo/hyperglycemia, or change in health status
Annually	• Lipids: Age 20-39 consider statin if additional risk factors (RF). Age 40-75: moderate-intensity statin for all DM pts regardless of lipid panel. If ≥ 1 ASCVD RF, then high intensity statin w/ target LDL-C < 70. Pts with clinical ASCVD (ACS, TIA), target LDL-C < 55 with statins + other agents • Urine mAlb/Cr, Cr/eGFR; ACEi/ARB if hypertensive w/ proteinuria <u>or</u> GFR < 60; refer to renal if GFR < 30 • Neuropathy exam: 10g monofilament (+ if no sensation at 4/10 sites, see PCOI); pinprick, vibration, or reflexes • Retinopathy screen w/ dilated eye exam or retinal photography; can consider q2-3y if normal exam(s) • LFTs: consider elastography and/or hepatology referral if elevated to evaluate for MASLD • Other: B12 for those on metformin. T1DM – yearly TSH.
Vaccines	• Influenza annually, COVID-19, HBV series if age < 60 and not immune, Zoster (age > 50), TDAP, Pneumococcus, RSV (CDC)

Basal Insulin Management (Diabetes Care 2026:49:S183-215)	
Criteria for initiation	• Consider if A1c $\geq 10\%$, random BG ≥ 300, fasting BG ≥ 250, or symptomatic; suspicion for T1DM or LADA/insulin deficiency; < 65yo on two agents with A1c $> 8\%$ (or ≥ 65yo and A1c $> 8.5\%$) on two occasions > 3mo apart; or A1c rising quickly. • Able to perform self-monitoring with glucometer; consider referral to DM educator
Initial dose	• Starting dose: A1c $< 8\%$: 0.1-0.2 U/kg/d, A1c $> 8\%$: 0.2-0.3 U/kg/d • Choice of agent: choose long-acting (glargine/detemir/degludec qd), preferred over intermediate-acting (NPH BID) to reduce risk of hypoglycemia • Route: pen (easier to use, more expensive) vs needle/syringe
Titration	• Increase by 2-4U or 10-15% q2-3d until AM fasting BS is 80-130 without hypoglycemia; savvy patients can self-titrate. • If hypoglycemia occurs or FPG < 80 without clear reason, decrease dose by 10-20% or 4U, whichever is greater.
If patients required additional control beyond basal insulin, the ADA recommends considering a GLP-1 receptor agonist or dual GIP/GLP-1 RA before initiation prandial insulin to minimize the risks of hypoglycemia and weight gain associated with insulin therapy.	

Prandial Insulin Management	
Criteria	• Consider if A1c still not at goal with basal insulin > 0.5U/kg/d and fasting glucose within target range (80-130).
Initial dose	• Strategy 1: add 1 rapid-acting insulin before largest meal $\rightarrow$ start w/ 4U or 0.1U/kg or 10% basal dose • Strategy 2: change to mixed insulin (e.g. fixed 70/30, NPH + regular) BID (before breakfast and dinner). Divide current basal dose into 2/3 AM, 1/3 PM or 1/2 AM, 1/2 PM. Confirm w/ pt they eat 3 meals regularly if choosing this.
Titration	• Increase dose by 1-2U or 10-15% q3d until target glucose reached (pre-prandial: 80-130; 1-2h post-prandial < 180) • If A1c still not controlled: add rapid-acting insulin to another meal and titrate as above • If hypoglycemia occurs or FPG < 80 without clear reason, decrease dose by 10-20% or 4U, whichever is greater

INSULIN SUPPLIES

- **Needles:** Pen needles for pts on insulin pen or 2) needle+ syringe 32G 4mm is less painful (higher gauge = thinner and shorter needle), but larger patients and high insulin doses often require longer/wider needle.
- **Syringes:** boxes of 100. Choose smallest syringe that will hold the dose (smaller barrel $\rightarrow$ clearer scale markings)
- **Alcohol swabs** (or patients can wash hands/skin with soap and water)
- **Glucometer & test strips:** Many choices, each with own strip brand. Most test strips come in boxes of 50-100. Some glucometers can read BG aloud (helpful for visually impaired). Tip: MassHealth only covers FreeStyle. Otherwise OneTouch Verio usually covered. Consider CGM as able.
- **** All durable medical equipment including test strips and glucometers requires an ICD-10 code on the script itself****

Use this barrel size...	With this dose range...
3/10 mL	30U or less
1/2 mL	31-50U
1 mL	51-100U